Econ 130_Information sheet for exam two (v.2) edits are in green

Exam #2: Thursday, May15th

Extra office hours (zoom): There will be something on Sunday (5/11 from 11:00 AM to 12:30 PM), and then some additional hours next week (something Tuesday and Wednesday).

- As always, exams are open book/open notes. Calculator is allowed, but not other electronic device (for example, iPads, smartwatches, etc are not allowed)
- Bring UCLA Photo ID

Topics on Exam:

I. Income tax topics.

A. Optimal income tax. Buildup part on Utility function assumption, MU, R curve (Laffer curve), MR. Deriving the MU/MR curve with respect to t and why it is upward slopping. Comparison between MU/MR for high and low income households, why one is above the other and its implications for optimal income tax rate across income groups.

B. The other aspect of this topic, of more importance, was the discussion on tax tables and changes in average rate for a given marginal rate, and changes in marginal rate for a given average rate—use of BLs/ICs to analyze how these changes affect leisure and tax revenue.

II. Public goods topics. Definition of public good, free-rider problem. Finding efficient quantity of public good, finding individual quantities desired, finding actual quantities (Median Voter Model), relevance of mean vs median consumer. Tiebout model. Lindahl model/pricing. Vickrey-Clarke-Groves mechanism as PRESENTED in the notes will not be on the exam; however, I gave a simple version (not in notes) in class this past Thursday (see lecture of May 8th, Thur for details), The section in the notes on property taxes (covered Thur—see lecture for details) is NOT on the exam.

See SQ#2 for relevant questions and see also the sample exam questions below. Skip question 3 and 4 and 5 on public goods on SQ#2.

SAMPLE TEST QUESTIONS

1. Optimal income taxation. Which of the following statements is FALSE?

a. Our basic model of optimal income tax rates assume that both high and low income face the same utility function.

b. For a given tax rate, we expect $U_{rich} > U_{poor}$ and $MU_{rich} < MU_{poor}$.

c. We expect the graph of the ratio (MU_i/MR_i) to be upward slopping.

d. If we were to compare the (MU_{rich}/MR_{rich}) curve with the (MU_{poor}/MR_{poor}) curve, we would find that (MU_{rich}/MR_{rich}) lies above (MU_{poor}/MR_{poor}) at any given tax rate (t).

e. At low tax rates, we would find that as t is raised a bit higher that R would rise and MR would fall.

f. if you believe two of the above statements are false, choose this option.

g. if you believe that all of the above are true statements, choose this option.

2. Optimal income taxation. Fill in the blanks/circle. Assume (MU_{high}/MR_{high}) =10 < (MU_{low}/MR_{low}) =18, then if the goal of the government is to maximize aggregate utility for a given income tax revenue, it should circle one (raise t on high income/lower t on high income/not change t on high income). If the government changes t optimally for low and high income, so as to gather one more dollar in tax revenue from one group but collect one less dollar from the other group, then would should see aggregate utility increasing by ______________ (numerical answer).

3.LABOR-LEISURE MODEL: INCOME TAX

a. The pre-tax wage is $1/hour, non-labor income is zero and the worker's time endowment is 24 hours per day. Pre-tax income up to $4 is not taxed, but all income greater than $4 is taxed at a rate of 10%. Under these circumstances the worker chooses to work 12 hours per day. On a clearly drawn graph sketch in the pre-tax budget, the after-tax budget line and the relevant indifference curve. What is the amount of tax revenue collected?

b. Suppose the government changes the tax rate such that any income above $4 is now taxed at a 20% rate. (so it is still the case that the first $4 of income is not taxed). Suppose the worker ends up choosing to work a number of hours such that the tax revenue collected is the same as in part (a). What are hours worked? Show this on the *same* graph as used for part (a). To get this outcome, what must we assume about the substitution and income effects?

c. Is it possible that the increase in the tax rate (without changing the cutoff income) could have resulted in no change in hours of work? *BRIEFLY* explain.

d. Return to part (a). Suppose the government implements the following new tax system: **ALL** income is taxed at a rate of t*. The tax rate t* is selected so as to hold tax revenue constant at 12 hours of work. Find the new tax rate t*. Does the worker increase or decrease hours of work? Use a graph to justify answer and briefly explain . Does the government collect more or less tax revenue from this worker? Again, justify answer with graph. Since we are by assumption holding tax revenue constant at 12 hours work, then using tax revenue = $(t^*)(W)(H-H^*) = (t^*)(W)(H)$ since H^* is zero (we are taxing all income).

4. True or false. Public goods.

_______To get the market demand (or marginal social benefit curve) for a public good we vertically add the individual demand curves (or individual marginal benefit curves). The reason for vertically adding is due to the non-rivalry characteristic of the public good.

__________If the *mean* quantity desired of a public good exceeds the *median* quantity, this would imply the actual quantity produced is less than the efficient quantity.

__________Assume three kinds of consumers/users of a public good: a high demander, a middle demander, and a low demander. If there was an increase in the high demander's demand, then other things constant this would make it more likely there will be under provision of the good.

5.Using the following information for the *next two* questions. Assume three consumers (3 residents or voters in a city) with the following MB_i functions (inverse demand functions):
MB_A= 180- (1/3)q_A, MB_B = 170 – (1/3)q_B, MB_c = 160 – (1/3)q_C,
Marginal cost of public good (C) = 210. The median voter model is assumed to apply for these questions. Fill in the blanks. The efficient amount of the public good is _____________ and the actual amount produced, assuming a median voter model, is ________________ .

6. Continue to use the above equations. Which of the following statements about the median voter model is FALSE?

a. The median voter quantity equals the mean voter quantity in this case.
b. If initially candidate #1 was to offer to produce 310 units if elected, and candidate #2 was to offer to produce 295 units if elected, then candidate #1 would get voter A's vote and candidate #2 would get the votes of B and C.
c. If initially candidate #1 was to offer to produce 310 units if elected, and candidate #2 was to offer to produce 295 units if elected, then candidate #1 would get voter A's vote and candidate #2 would get the votes of B and C; but candidate #1 could capture B's vote by instead offering 316 units. d. competition among candidates for votes would drive the equilibrium offer to 300.
e. If A's demand for the public good increased from MB_A= 180- (1/3)q_A to MB_A= 270- (1/3)q_A, this would change the efficient quantity of the public good, but it would not change the actual amount produced.
f. At the actual amount of the public good produced, voter C's desired amount would be less than the amount actually produced.
g. If you believe two of the above are false, choose this option.
h. If you believe three of the above are false choose this option.
i. If you believe that all of the above are true, choose this option.

7. Tiebout model. Suppose there are two cities Z and V. City Z has three voters, two of type of A, MB_A= 180- (1/3)q_A ,and one of type C, MB_C = 160 – (1/6)q_C. For City V, three voters, one of type A, and two of type C. Cost is $210 and is evenly divided among the residents. In this case, the Tiebout model would predict:

a. Initially, in city Z: $q_{efficient}$ = 310 and q_{actual} = 330.
b. Initially, in city V: $q_{efficient}$ = 300 and q_{actual} = 270.
c. After any migration across cities, in city Z: $q_{efficient}$ = 300 and q_{actual} = 330.
d. After any migration across cities, in city V: $q_{efficient}$ = 310 and q_{actual} = 270.
e. Type C could increase consumer surplus by migrating from city V to city Z.
f. Type A could increase consumer surplus by migrating from city Z to city V.
g. If you believe two of the above are correct, choose this option.
h. If you believe that all of the above are correct, choose this option.

~~9. Fill in the blanks. Assume $MB_A = 180 - (1/3)q_A$, $MB_B = 170 - (1/3)q_B$, $MB_c = 160 - (1/3)q_C$, Marginal cost of public good (C) = 210. Assume that the cost of providing the public good is financed by property taxes.~~

~~The equation for aggregate rents = ____.~~

~~The equation for aggregate property taxes = ____.~~

~~The equation for aggregate property values = ____.~~

~~The quanity of public good that maximizes aggregate property value is ____________________.~~

8. Unanimity Rules and Lindahl pricing. . Assume two users of a public good, A & B, with $MB_A = 110 - Q_A$ and $MB_B = 90 - Q_B$. Assume c = 106. with S_A being A's *share* of the cost (that is, the tax or price imposed on A expressed as %) and S_B being B's share of the cost. a. What is the efficient Q? ____________________________.

b. Suppose we start with 50% share for A. Fill in row one of the table below. If they disagree on quantities, suppose the government changes the share, raising it on A to, say, 0.55. Fill in row two. Suppose row three is equilibrium outcome, fill in row three (three decimal places for shares).

S_A (in % terms)	**S_B (in % terms)**	**c**	**S_A*c**	**S_B*c**	**A's desired amount**	**B's desired amount**
0.5	0.5					
0.55	0.45					

c. Draw a graph showing the Lindal prices.

9. ADDED question.

Modified/Simplfied version of VCG mechanism. Suppose there is one user of a public good—person Z. The "true value" this person places on the good is \$100. A government official makes the following offer to Z: "We will produce the good if your reported value $v_z^* > X$ where X is some unknown number and we will charge you X; if however $v_z^* < X$ we will not produce it". What value will Z report? Is their an incentive to lie about value and underreport it?

~~9. Vickrey-Clarke-Groves mechanism. Suppose there are two people in this economy. The marginal benefit schedules for some pure public good are given by $MB_1 = 9 - Q_1$ for individual one and $MB_2 = 5 - Q_2$ for individual two. The marginal cost of providing the good is constant at \$6.~~

a. ~~What is consumer 1's net total gain (U_1) under the Vickrey-Clarke-Groves mechanism assuming both consumers report their MB schedules truthfully?~~

b. ~~Suppose consumer 1 now "lies" (& consumer 2 reports truthfully) by reporting a MB schedule of $MB_1^* = 7 - Q_1$. Demonstrate that consumer 1 does not have an incentive to do so under a ClarkeGroves Tax.~~

SAMPLE TEST QUESTIONS--Answers

1. Optimal income taxation. Which of the following statements is FALSE?

a. Our basic model of optimal income tax rates assume that both high and low income face the same utility function.
b. For a given tax rate, we expect $U_{rich} > U_{poor}$ and $MU_{rich} < MU_{poor}$.
c. We expect the graph of the ratio (MU_i/MR_i) to be upward sloping.
d. If we were to compare the (MU_{rich}/MR_{rich}) curve with the (MU_{poor}/MR_{poor}) curve, we would find that (MU_{rich}/MR_{rich}) lies above (MU_{poor}/MR_{poor}) at any given tax rate (t). False.
e. At low tax rates, we would find that as t is raised a bit higher that R would rise and MR would fall.
f. if you believe two of the above statements are false, choose this option.
g. if you believe that all of the above are true statements, choose this option.

2. Optimal income taxation. Fill in the blanks/circle. Assume (MU_{high}/MR_{high}) =10 < (MU_{low}/MR_{low}) =18, then if the goal of the government is to maximize aggregate utility for a given income tax revenue, it should (*circle one:* raise t on high income/lower t on high income/not change t on high income). If the government changes t optimally for low and high income, so as to gather one more dollar in tax revenue from one group but collect one less dollar from the other group, then would should see aggregate utility increasing by ____8__________ (numerical answer).

3.LABOR-LEISURE MODEL: INCOME TAX

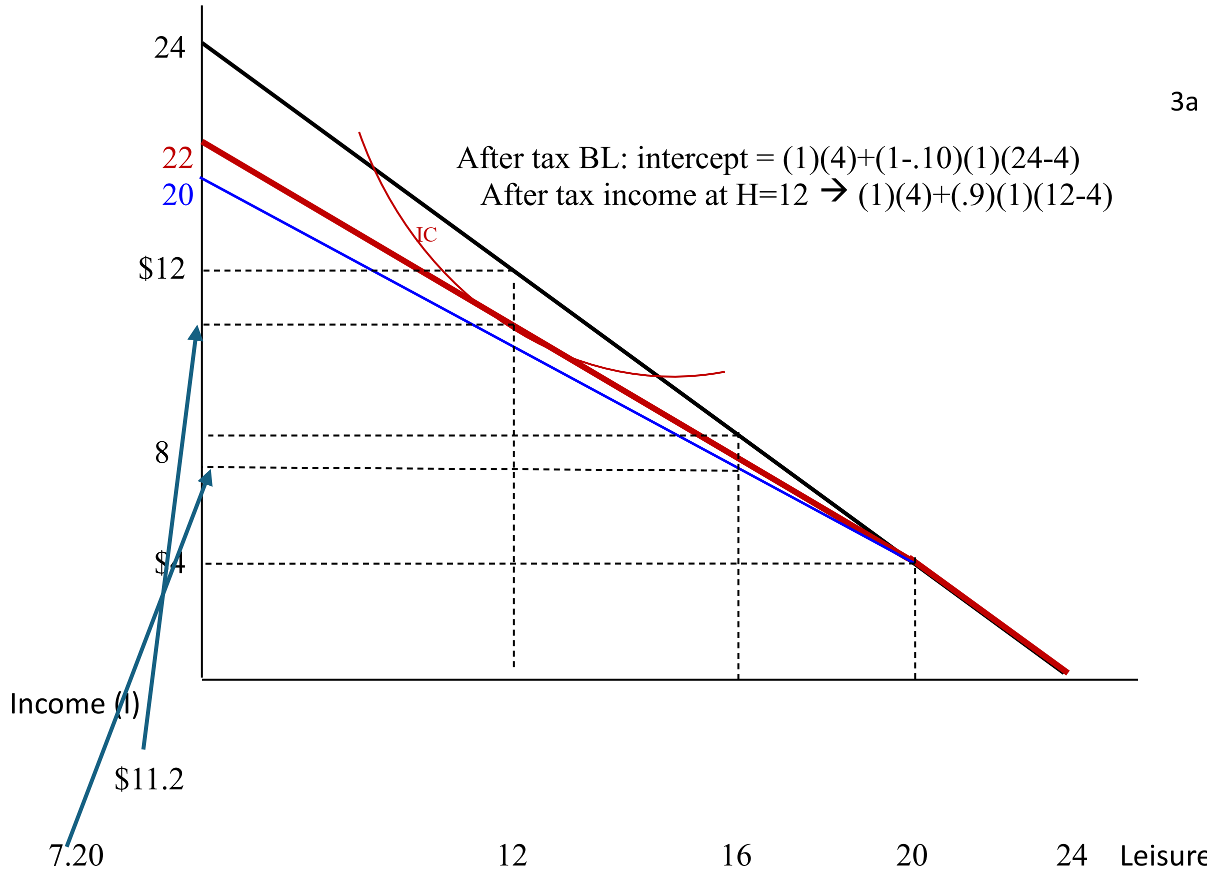
a. The pre-tax wage is $1/hour, non-labor income is zero and the worker's time endowment is 24 hours per day. Pre-tax income up to $4 is not taxed, but all income greater than $4 is taxed at a rate of 10%. Under these circumstances the worker chooses to work 12 hours per day. On a clearly drawn graph sketch in the pre-tax budget, the after-tax budget line and the relevant indifference curve. What is the amount of tax revenue collected?
for the numerical part, taxes paid = t*W*(H – H*) = (.10)(1)(12-4) = $0.80

b. Suppose the government changes the tax rate such that any income above $4 is now taxed at a 20% rate. (so it is still the case that the first $4 of income is not taxed). Suppose the worker ends up choosing to work a number of hours such that the tax revenue collected is the same as in part (a). What are hours worked? Show this on the *same* graph as used for part (a). To get this outcome, what must we assume about the substitution and income effects? for the numerical part, taxes paid = 0.80 = t*W*(H – H*) = (.20)(1)(H-4) = $0.80 H = 8. the after-tax wage in the relevant region declined. By lowering the price of leisure this would increase L and decrease H (SE). However, this tax change makes the person less wealthy (IE) which would decrease L and increase H. Since the result above shows that H declined from 12 to 8, this implies the SE > IE.

c. Is it possible that the increase in the tax rate (without changing the cutoff income) could have resulted in no change in hours of work? *BRIEFLY* explain.
Yes, if the SE = IE.

d. Return to part (a). Suppose the government implements the following new tax system: **ALL** income is taxed at a rate of t*. The tax rate t* is selected so as to hold tax revenue constant at 12 hours of work. Find the new tax rate t*. Does the worker increase or decrease hours of work? Use a graph to justify answer and briefly explain . Does the government collect more or less tax revenue from this worker? Again, justify answer with graph. Since we are by assumption holding tax revenue constant at 12 hours work, then using tax revenue = $(t^*)(W)(H-H^*) = (t^*)(W)(H)$ since H* is zero (we are taxing all income). Plugging in, $0.80 = (t^*)(W)(H) = (t^*)(\$1)(12) \rightarrow t^* = 0.80/12$ or $8/120 = 0.067$ (6.7%).

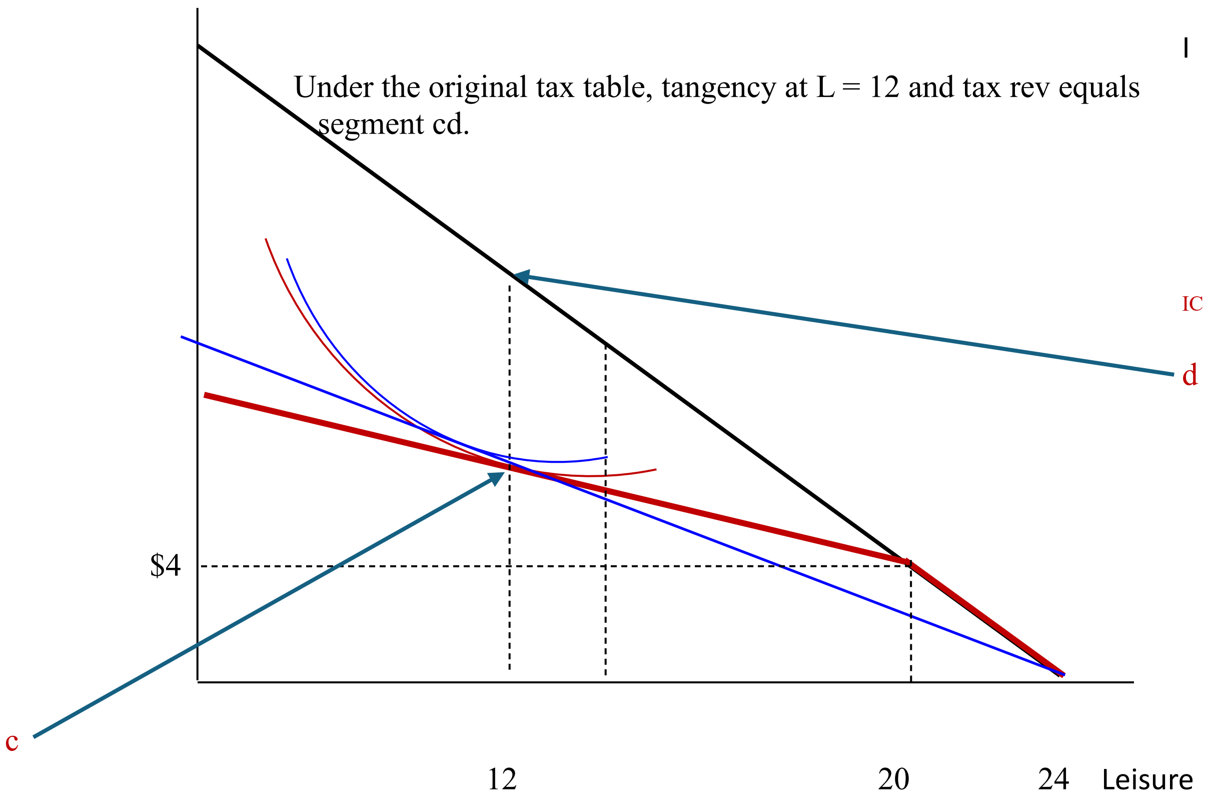
3a & b.



Intercept for blue BL (t = 20%): (1)(4) +(1-.20)(1)(24-4)= 20

At H = 8, L=16, then pre-tax income is $8, after-tax income = (1)(4)+ (.8)(1)(8-4)

3d.



Since tax rate lower under flat-tax (6.7%), the after tax wage, or slope of the new budget line (blue) is flatter. Since by assumption we are holding tax revenue constant AT H-12, it follows that the blue BL must go through point c. It is NOT tangent at that point, rather intersects at that point. Notice there necessarily exists bundles (for L < 12) that are on higher IC and are therefore preferred (see blue IC).Thus, L falls and H rises. Tax revenue also rises since the vertical gap at this new point must be larger.

4.

____True____To get the market demand (or marginal social benefit curve) for a public good we vertically add the individual demand curves (or individual marginal benefit curves). The reason for vertically adding is due to the non-rivalry characteristic of the public good.

___False_______If the *mean* quantity desired of a public good is below the *median* quantity, this would imply the actual quantity produced is less than the efficient quantity.

_True___Assume three kinds of consumers/users of a public good: a high demander, a middle demander, and a low demander. If there was an increase in the high demander's demand, then other things constant this would make it more likely there will be under provision of the good.

5.Using the following information for the *next two* questions. Assume three consumers (3 residents or voters in a city) with the following MB_i functions (inverse demand functions):

MB_A= 180- (1/3)q_A, MB_B = 170 – (1/3)q_B, MB_c = 160 – (1/3)q_C,

Marginal cost of public good (C) = 210. The median voter model is assumed to apply for these questions. Fill in the blanks. The efficient amount of the public good is ___300____and the actual amount produced, assuming a median voter model, is ______300____________ .

MSB= 510- q = 210 or $Q_{efficient}$ = 300 Desired quantities:

MB_A= 180- (1/3)q_A = 70 → q_A= 330

MB_B = 170 – (1/3)q_B = 70 → q_B= 300 MB_c = 160 – (1/3)q_C = 70 → q_A= 270

6. Continue to use the above equations. Which of the following statements about the median voter model is FALSE?

a. The median voter quantity equals the mean voter quantity in this case.
b. If initially candidate #1 was to offer to produce 310 units if elected, and candidate #2 was to offer to produce 295 units if elected, then candidate #1 would get voter A's vote and candidate #2 would get the votes of B and C.
c. If initially candidate #1 was to offer to produce 310 units if elected, and candidate #2 was to offer to produce 295 units if elected, then candidate #1 would get voter A's vote and candidate #2 would get the votes of B and C; but candidate #1 could capture B's vote by instead offering 316 units .
d. competition among candidates for votes would drive the equilibrium offer to 300.
e. If A's demand for the public good increased from MB_A= 180- (1/3)q_A to MB_A= 270- (1/3)q_A, this would change the efficient quantity of the public good, but it would not change the actual amount produced.
f. At the actual amount of the public good produced, voter C's desired amount would be less than the amount actually produced.
g. If you believe two of the above are false, choose this option.
h. If you believe three of the above are false choose this option.
i. If you believe that all of the above are true, choose this option.

7. Tiebout model. Suppose there are two cities Z and V. City Z has three voters, two of type of A, $MB_A = 180 - (1/3)q_A$,and one of type C, $MB_C = 160 - (1/6)q_C$. For City V, three voters, one of type A, and two of type C. Cost is $210 and is evenly divided among the residents. In this case, the Tiebout model would predict:

a. Initially, in city Z: $q_{efficient} = 310$ and $q_{actual} = 330$.
b. Initially, in city V: $q_{efficient} = 300$ and $q_{actual} = 270$.
c. After any migration across cities, in city Z: $q_{efficient} = 300$ and $q_{actual} = 330$.
d. After any migration across cities, in city V: $q_{efficient} = 310$ and $q_{actual} = 270$.
e. Type C could increase consumer surplus by migrating from city V to city Z.
f. Type A could increase consumer surplus by migrating from city Z to city V.
g. If you believe two of the above are correct, choose this option.
h. If you believe that all of the above are correct, choose this option.

- Initially, in City Z: MSB= [180-(1/3)q] + [180-(1/3)q] + [160-(1/3)q] = 520 – q

520 – q = 210 or q*= 310
Median voter is A, and A's desired amount is $180 - (1/3)q_A = 70$ or $q_A = 330$.

- Initially, in City V: MSB= [180-(1/3)q] + [160-(1/3)q] + [160-(1/3)q] = 500 – q

500 – q = 210 or q*= 290
Median voter is C, and C's desired amount is $160 - (1/3)q_A = 70$ or $q_c = 270$

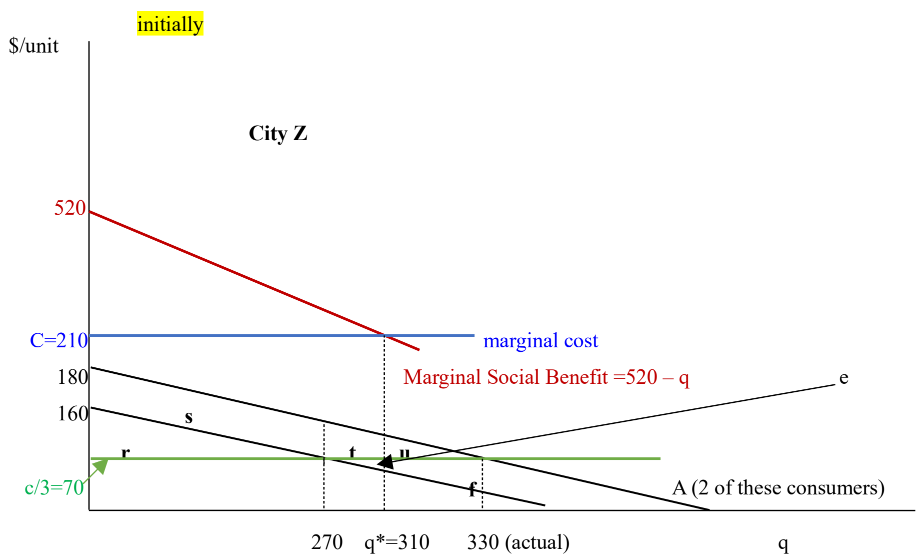
- Type C in Z would move to V, and Type A in V would move to Z, so after this migration:
- After, in City Z: MSB= [180-(1/3)q] + [180-(1/3)q] + [180-(1/3)q] = 540 – q

540 – q = 210 or q*= 330
Median voter is A, and A's desired amount is $180 - (1/3)q_A = 70$ or $q_A = 330$.

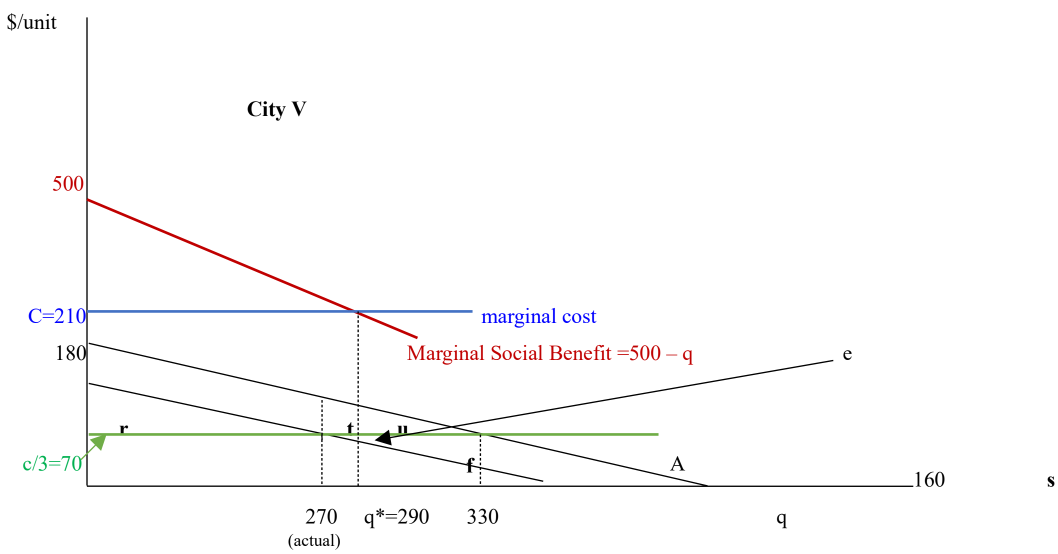
- After, in City V: MSB= [160-(1/3)q] + [160-(1/3)q] + [160-(1/3)q] = 480 – q

480 – q = 210 or q*= 270
Median voter is C, and C's desired amount is $160 - (1/3)q_A = 70$ or $q_c = 270$

Consumer surplus to person C under this outcome = r – ef (I assuming this is, on net, positive).
Consumer surplus to each A under this outcome = rstu.



City V:
Consumer surplus to C under this outcome = r
Consumer surplus to each A under this outcome = rs.



If voter C located in City Z moved to City V then C's CS would change from *(r-ef)* to *r*, a gain of ef → ΔCS_C = +ef
If voter A in City V moved to City Z, A's consumer surplus would increase from *rs* to *rstu* →ΔCS_A = +tu

~~9. Fill in the blanks. Assume MB_A= 180- (1/3)q_A, MB_B = 170 – (1/3)q_B, MB_c = 160 – (1/3)q_C, Marginal cost of public good (C) = 210. Assume that the cost of providing the public good is financed by property taxes.~~

~~The equation for aggregate rents =~~ ____.

~~The equation for aggregate property taxes =~~ ____.

~~The equation for aggregate property values =~~ ____.

~~The quanity of public good that maximizes aggregate property value is~~ ____________________.

8. Unanimity Rules and Lindahl pricing. . Assume two users of a public good, A & B, with MB_A = 110 – Q_A and MB_B = 90 - Q_B. Assume c = 106. with S_A being A's *share* of the cost (that is, the tax or price imposed on A expressed as %) and S_B being B's share of the cost. a. What is the efficient Q? _________47___.   200-2q=106→ q=47

b. Suppose we start with 50% share for A. Fill in row one of the table below. If they disagree on quantities, suppose the government changes the share, raising it on A to, say, 0.55. Fill in row two. Suppose row three is equilibrium outcome, fill in row three (three decimal places for shares).

S_A (in % terms)	**S_B (in % terms)**	**c**	**S_A*c**	**S_B*c**	**A's desired amount**	**B's desired amount**
0.5	0.5	106	53	53	57	37
0.55	0.45	106	58.3	47.7	51.7	42.3
.594	.406	106	~63	~43	47	47

c. Draw a graph showing the Lindal prices.

Row 1: 110-q_a=53→ 57 and 90-q_b = 53→ 37 No agreement.
Row 2: 110-q_a=58.3→ 51.7 and 90-q_b =47.7→42.3. No agreement.
Row 3: the efficient quantity is 47, so
MB_A = 110 – Q_A = 110-47= 63 and MB_B = 90 - Q_B= 90-47=43
MSB at the efficient quantity is 200-2q= 200-2(47)=106
S_i= MB_i/MSB -> for A: 63/106=.594 and for B: 43/106=0.406

Answer to 9. Z will report true value of \$100. If \$100 > x it gets produced, Z pays X and gets a net gain of Z-X; if \$100 < x, then not produced but Z is happy it is not produces since would have been charged more than values it at.

Z has no incentive to under-report: Under-report means $v_z^{true} > v_z^*$. Two scenarios with this ordering. First, there is $v_z^{true} > v_z^* > x$ ---> the good is produced whether under-report or not and would get same net gain of $v_z^{true} – x$ either way, thus no advantage to under-reporting in this scenario. Second scenario, $v_z^{true} > x > v_z^*$ → this case, it is NOT produced since report a value less than x; but by under-reporting you hurt yourself by losing out on the net gain of $v_z^{true} – x$. Thus, under-reporting would be your disadvantage. Since under-reporting never leads to any advantage and only opens the door to a bad outcome for Z , it follows that it never pays to under report. Should report your true value!

~~9. Vickrey-Clarke-Groves mechanism. Suppose there are two people in this economy. The marginal benefit schedules for some pure public good are given by $MB_1 = 9 – Q_1$ for individual one and $MB_2= 5 -Q_2$ for individual two. The marginal cost of providing the good is constant at \$6.~~

a. ~~What is consumer 1's net total gain (U_1) under the Vickrey-Clarke-Groves mechanism assuming both consumers report their MB schedules truthfully?~~

b. ~~Suppose consumer 1 now "lies" (& consumer 2 reports truthfully) by reporting a MB schedule of $MB_1^*= 7 – Q_1$. Demonstrate that consumer 1 does not have an incentive to do so under a ClarkeGroves Tax.~~

~~-find q produced if truthful reporting: 14 –2Q = 6 or Q = 4.~~

~~– figure out the Clarke-Groves tax → The Clarke-Groves Tax is given by~~

~~$T_1 = TC – TB_2 = (\$6)(4) – [5Q-0.5Q^2] = \$24 – [5(4) – 0.5(4)^2] = 24-20+8= +\$12$~~

~~– Third, figure out *net* total benefits →~~

~~$U_1 = TB_1 – T_1 = 9Q-0.5Q^2 – 12= 9(4)-0.5(4)^2 – 12 = 36-8 -12= \$16$~~

~~-find q produced if consumer 1 does not tell truth: 12 –2Q = 6 or Q = 3.~~

~~– figure out the Clarke-Groves tax → The Clarke-Groves Tax is given by~~

~~$T_1 = TC – TB_2 = (\$6)(3) – [5Q-0.5Q^2] = \$18 – [5(3) – 0.5(3)^2] = 18-15+4.50= 7.50$~~

~~– Third, figure out *net* total benefits →~~

~~$U_1 = TB_1 – T_1 = 9Q-0.5Q^2 – 7.50= 9(3)- 0.5(3)^2 – 7.50= 27- 4.50 – 7.50= \$15$. Net gains are lower by not telling the truth.~~